UPDATED SYLLABUS

SPECIALISED PYQ

THERMODYNAMICS

60 MINUTES

40 QUESTIONS

Q1. Consider the following reaction:
$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$ $\Delta_rH^\circ = -483.64$ kJ
What is the enthalpy change for decomposition of one mole of water? (Choose the right option). **(2023)**
(a) 120.9 kJ
(b) 241.82 kJ
(c) 18 kJ
(d) 100 kJ

Q2. The equilibrium concentrations of the species in the reaction $A + B \rightleftharpoons C + D$ are 2, 3, 10 and 6 mol L^{-1}, respectively at 300 K. ΔG° for the reaction is: (R = 2 cal/mol K) **(2023)**

(a) −137.26 cal
(b) −1381.80 cal
(c) −13.73 cal
(d) 1372.60 cal

Q3. Which amongst the following options is the correct relation between change in enthalpy and change in internal energy? **(2023)**

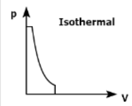
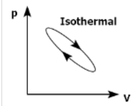
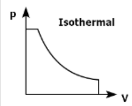
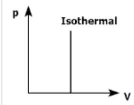
(a) $\Delta H = \Delta U + \Delta n_gRT$
(b) $\Delta H - \Delta U = -\Delta nRT$
(c) $\Delta H + \Delta U = \Delta nR$
(d) $\Delta H = \Delta U - \Delta n_gRT$

Q4. One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume. ΔU for this process is: **(2022)**

(Use R = 8.314 J k^{-1} mol^{-1})
(a) 0 J
(b) 1260 J
(c) 2520 J
(d) 5040 J

Q5. A vessel contains 3.2 g of dioxygen gas at STP (273.15 K and 1 atm pressure). The gas is now transferred to another vessel at constant temperature, where pressure becomes one third of the original pressure. The volume of new vessel in L is: **(2022)**
(Given: molar volume at STP is 22.4 L)
(a) 67.2
(b) 6.72
(c) 2.24
(d) 22.4

Q6. Which of the following p-V curve represents maximum work done? **(2022)**

1.		2.	
3.		4.	

Q7. Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas? **(2021)**
(a) CP – CV = R
(b) CP = RCV
(c) CV = RCP
(d) $C_P + C_V = R$

Q8. For irreversible expansion of an ideal gas under isothermal condition, the correct option is: **(2021)**

(a) $\Delta rH>0$ *and* $\Delta rS<0$
(b) $\Delta rH<0$ *and* $\Delta rS<0$
(c) $\Delta rH<0$ *and* $\Delta rS<0$
(d) $\Delta rH>0$ *and* $\Delta rS>0$

Q9. The correct option for free expansion of an ideal gas under adiabatic condition is: **(2020)**

(a) $q=0, \Delta T<0$ *and* $w>0$
(b) $q<0, \Delta T=0$ *and* $w=0$
(c) $q>0, \Delta T>0$ *and* $w>0$
(d) $q=0, \Delta T=0$ *and* $w=0$

Q10. If for a certain reaction ΔrH is 30 kJ mol^{-1} at 450 K, the value of $\Delta_r S$ (in JK^{-1} mol^{-1}) for which the same reaction will be spontaneous at the same temperature is **(2020 Covid Re-NEET)**

(a) –33
(b) 33
(c) –70
(d) 70

Q11. At standard conditions, if the change in the enthalpy for the following reaction is –109 kJ mol^{-1}

$H2(g)+Br2(g)\rightarrow 2HBr(g)$
Given that bond energy of H_2 and Br_2 is 435 kJ mol^{-1} and 192 kJ mol^{-1}, respectively, what is the bond energy (in kJ mol^{-1}) of HBr? **(2020 Covid Re-NEET)**
(a) 736
(b) 518
(c) 259

(d) 368

Q12. Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is (Give that 1 L bar = 100 J) **(2019)**

(a) –30 J
(b) 5 kJ
(c) 25 J
(d) 30 J

Q13. In which case change in entropy is negative? **(2019)**

(a) Evaporation of water
(b) Expansion of a gas at constant temperature
(c) Sublimation of solid to gas
(d) $2H(g)\rightarrow H2(g)$

Q14. The bond dissociation energies of X_2, Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is –200 kJ mol^{-1}.
(c) $-8.314\ J\ mol-1\ K-1\times 300\ K\times \ln(4\times 1013)$
(d) $-8.314\ J\ mol_{-1}\ K_{-1}\times 300\ K\times lb(2\times 10_{13})$

Q15. The bond dissociation energy of X_2 will be **(2018)**
(a) 200 kJ mol–1
(b) 100 kJ mol–1
(c) 400 kJ mol–1
(d) 800 kJ mol–1

Q16.A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be: **(2017-Delhi)**

(a) +505 J
(b) 1136.25 J
(c) –500 J
(d) –505 J

Q17. For a given reaction, $\Delta H=35.5$ KJ mol_{-1} and $\Delta S=83.6$ $JK_{-1}mol_{-1}$. The reaction is spontaneous at: (Assume that ΔH and ΔS do not vary with temperature) **(2017-Delhi)**

(a) T > 298 K
(b) T < 425 K
(c) T > 425 K
(d) All temperatures

Q18. Of the following, the largest value of entropy at 25°C and 1 atm is that of : **(2017-Gujarat)**

(a) CH_4
(b) H_2
(c) C_2H_6
(d) C_2H_2

Q19. Under isothermal and reversible conditions, the term "free energy" in thermodynamics signifies : **(2017-Gujarat)**

(a) Expansion work done on the system
(b) Non-expansion work done by the system
(c) Expansion work done by the system
(d) Non expansion work done on the system

Q20. For a sample of perfect gas when its pressure is changed isothermally from P_i to P_f, the entropy change is given by: **(2016-I)**

(a) $\Delta S = nRT\ ln\left(\frac{P_f}{P_i}\right)$
(b) $\Delta S = nRT\ ln\left(\frac{P_i}{P_f}\right)$
(c) $\Delta S = nR\ ln\left(\frac{P_f}{P_i}\right)$
(d) $\Delta S = nR\ ln\left(\frac{P_i}{P_f}\right)$

Q21. The correct thermodynamic conditions for the spontaneous reaction at all temperatures is: **(2016-II)**
(a) $\Delta H < 0$ and $\Delta S < 0$
(b) $\Delta H < 0$ and $\Delta S = 0$
(c) $\Delta H > 0$ and $\Delta S < 0$
(d) $\Delta H < 0$ and $\Delta S > 0$

Q22. Consider the following liquid-vapour equilibrium.

Liquid $\rightleftharpoons$ Vapour . Which of the following relations is correct? **(2016-I)**
(a) $\frac{d\ln P}{dT^2} = \frac{-\Delta H_v}{T^2}$
(b) $\frac{d\ln P}{dT} = \frac{-\Delta H_v}{RT^2}$
(c) $\frac{d\ln G}{dT^2} = \frac{\Delta H_v}{RT^2}$
(d) $\frac{d\ln P}{dT} = \frac{-\Delta H_v}{RT}$

Q23. The heat of combustion of carbon to CO_2 is –393.5 kJ/mol. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is: **(2015)**

(a) +315 kJ
(b) –630 kJ
(c) –3.15 kJ
(d) –315 kJ

Q24. For the reaction, $X_2O_4(l) \rightarrow 2XO_2(g)$, $\Delta U = 2.1\ kcal$, $\Delta S = 20\ cal\ K^{-1}$ at 300 K. Hence, ΔG is : **(2014)**

(a) –2.7 kcal
(b) 9.3 kcal

(c) –9.3 kcal
(d) 2.7 kcal

ANSWER KEY

1. Ans. (b)
2. Ans. (b)
3. Ans. (a)
4. Ans. (a)
5. Ans. (b)
6. Ans. (b)
7. Ans. (a)
8. Ans. (b)
9. Ans. (c)
10. Ans. (d)
11. Ans. (d)
12. Ans. (d)
13. Ans. (a)
14. Ans. (d)
15. Ans. (d)
16. Ans. (d)
17. Ans. (c)
18. Ans. (c)
19. Ans. (b)
20. Ans. (d)
21. Ans. (d)
22. Ans. (b)
23. Ans. (a)
24. Ans. (a)

1. Ans.(b)
Decomposition for 1 mole of water
$H_2O(g) \longrightarrow H_2(g) + \frac{1}{2}O_2(g)$; $\Delta H = +\frac{483.64}{2}$
$\Delta H = +241.82$ kJ

2. Ans.(b)
$A + B \rightleftharpoons C + D$
$[A] = 2$ mol L^{-1}
$[B] = 3$ mol L^{-1}
$[C] = 10$ mol L^{-1}
$[D] = 6$ mol L^{-1}
$\Delta G° = -2.303\ RT \log K_{eq}$
$= -2.303\ RT \log \frac{[C][D]}{[A][B]}$
$= -2.303 \times 2 \times 300 \times \log \frac{10 \times 6}{2 \times 3}$
$= -2.303 \times 2 \times 300 \times \log 10$
$= -1381.8$ cal

3. Ans.(a) $\Delta H = \Delta U + \Delta n_g RT$

4. Ans.(a)
$\Delta U = nC_v\Delta T$
For isothermal condition; $\Delta T = 0$
$\therefore \Delta U = 0$

5. Ans.(b)
At constant temperature and amount
$P_1V_1 = P_2V_2$
$P_1V_1 = \frac{P_1}{3}V_2$ $[\therefore P_2 = \frac{P_1}{3}]$
$V_2 = 3V_1$
mole of $O_2(g) = 3.2/32 = 0.1$ mole
Volume of $O_2(g) = (0.1 \times 22.4)L = 2.24$ L
At STP (V_1)
$V_2 = 3V_1 = 3 \times 2.24 = 6.72$

6. Ans.(b)
Work done under any thermodynamic process can be determined by area under the 'p-V' graph. As it can be observed maximum area is covered in option '2'.

7. Ans.(a)
For one mole of an ideal gas $C_P - C_V = R$

8. Ans.(b)
For Irreversible expansion of an ideal gas under Isothermal condition
$\Delta U = 0$
$\Delta S_{Total} \neq 0$

9. Ans.(c)
Given reaction, $2Cl(g) \rightarrow Cl_2(g)$

The reaction is exothermic since a bond is being formed in the given reaction, i.e., ΔH is also –ve. So $\Delta_r H<0$ and
Entropy is decreasing (–ve) in the reaction as no. of molecules are decreasing which reduces the randomness of the system.
Thus, $\Delta r S<0$
So, (c) option is correct.

10. Ans.(d)
Free expansion, since $P_{ext}=0$
$w=-P_{ex}\Delta V=0$; So, work done = 0
For adiabatic condition, q = 0
From first law of thermodynamics
$\Delta E=q+w$
$\therefore \Delta E=0$
Internal energy of an ideal gas is a function of temperature
$\therefore$ If internal energy remains constant
$\therefore \Delta T=0$
Thus (d) option is correct.

11. Ans.(d)
For spontaneous reaction, $\Delta_r G$ must be less than zero
So, $\Delta r G=\Delta r H-T\Delta r S<0$
or, $\Delta_r S>\frac{\Delta_r H}{T}$
$>\frac{30,000}{450}=66.67\ JK^{-1}\,mol^{-1}$
For reaction to be spontaneous the value of $\Delta_r S$ must be greater than 66.67 J that is 70 JK^{-1} mol^{-1}

12. Ans.(d)
$H_2(g)+Br_2(g)\rightarrow 2HBr(g)$; $\Delta H=-109 kJ\ mol^{-1}$
ΔH= sum of bond energy of reactants – sum of bond energy of products
$(BE_{H-H})+(BE_{Br-Br})-2(BE_{H-Br})$ $-109=(435)+(192)-2(BE_{H-Br})$
$BE_{H-Br}=368\ kJ\ mol^{-1}$
S13. Ans.(a)
$W_{irr}=-P_{ext}\Delta V=-2\ bar\times(0.25-0.1)L$
= –0.30 L-bar = –0.30 × 100 J
⇒ –30 J

14. Ans.(d)
(i) For evaporation of water □S > 0;
□S = +ve
i In expansion of gas at constant T

□S > 0 ; □S = +ve
i S→g

□S = +ve
i 2Hg→H2(g),ΔS<0 (∵ □ng < 0)

No. of particle decreases from reactant to product side.

15. Ans.(d)
The reaction for $\Delta_f H°(XY)$
$12X_2(g)+12Y_2(g)\rightarrow XY(g)$
Bond energies of X_2, Y_2 and XY are x, $x2$, x respectively
∴ $\Delta H=(x2+x4)-x2=-200$
On solving, we get
⇒ $x4=-200$
⇒ x = 800 kJ/mole

16. Ans.(d)
□U = q + W
For adiabatic process, q = 0
∴ U = W = –P. V
= –2.5 atm × (4.50 – 2.50) L
= –2.5 × 2 L-atm ⇒ –5 × 101.3 J
= –506.5 J ⇒ –505 J

17. Ans.(c)
□G = □H – T□S
For spontaneous reaction, □G must be negative. For negative value of □G, □H should be less than T□S. It is possible when T > 425 K.
□H = 35.5 KJ/mol = 35500 J/mol
T□S = (425) (83.6) = 35530
T□S > □H
Hence, the reaction is spontaneous.

18. Ans.(c)
The molecule having more number of bonds have largest value of entropy.
∴ C_2H_6 have large value of entropy.

19. Ans.(b)
Under isothermal reversible conditions, the term “free-energy” in thermodynamics signifies that “No expansion work done by the system”.
$-\Delta G_{system}=W_{non-expansion}$

20. Ans.(d)
For isothermal process
$\Delta S=nRln_{PiPf}$

21. Ans.(d)
For any spontaneous process, □G = (–ve) and □S = (+ve) that is increase in entropy. So, at □H < 0 and □S > 0 at all temperatures according the reaction will be spontaneous.

22. Ans.(b)
$dlnPdT=-\Delta H_vRT2$ (this is Clausis Clapeyron equation).

23. Ans.(a)
$C(s)+O_2\rightarrow CO_2(g)$ $\Delta H=-393.5\ KJ/mole$
$\Delta Hr=\Delta HfCO2$

So heat of formation of CO_2
= 393.5 kJ/mole
or we can say
Heat released on formation of 44 gm CO_2 = 393.5 kJ
Heat released on formation of 35.2 gm of $CO_2 = \frac{393.5}{44} \times 35.2 = 314.8$
$\Rightarrow \simeq 315$ kJ

24. Ans.(a)
According + 2 × 2 × 300 = 3300
$\Delta G = \Delta H - T\Delta S = 3300 - 300 \times 20$
$= -2700$ cal $= -2.7$ kcal